

解方程

$$\frac{dy}{dx} = \frac{x - y + 1}{x + y^2 + 3}$$

化简为

$$(y - x - 1)dx + (x + y^2 + 3)dy = p dx + q dy = 0$$

观察到

$$\partial_x q = 1 = \partial_y p$$

那么可以令

$$du = p dx + q dy$$

$$\partial_x u = y - x - 1, \partial_y u = x + y^2 + 3$$

积分得到

$$u^{(x)} = xy - \frac{1}{2}x^2 - x + f(y), u^{(y)} = xy + \frac{1}{3}y^3 + 3y + g(x)$$

对比得到

$$f(y) = \frac{1}{3}y^3 - 3y, g(x) = \frac{1}{2}x^2 + x$$

那么

$$u = xy - \frac{1}{2}x^2 - x + \frac{1}{3}y^3 - 3y = C$$